Social Media Bot Detection - Presentation

Slide 1: Title Slide

Title: Social Media Bot Detection System
Subtitle: A Multi-Modal Machine Learning Approach
Course: Minor in Data Science

Speaker Notes:
Good morning everyone. Today I will be presenting my Data Science minor project: A Social Media Bot Detection System. I built a full-stack AI application designed to catch malicious bots hiding among real users.

Slide 2: The Problem

The Threat: Automated bots cause fraud, spread misinformation, and manipulate trends.
The Evolution: Bots are no longer just spamming; they use AI to generate realistic bios and 'Deepfake' profile pictures.
The Flaw: Traditional systems only look at numbers (like follower counts), which is no longer enough.

Speaker Notes:
The problem today is that bots are getting smarter. They don't just spam; they use AI to generate fake faces and realistic text. Standard detection systems that only look at follower counts are failing. We need a smarter solution.

Slide 3: The Goal & Solution

Goal: Build an end-to-end detection tool with a realistic UI.
The Solution: A Multi-Modal AI approach. Analyzing 3 Pillars of Data:
1. Behavioral Stats (Numbers)
2. Profile Bio (Text)
3. Profile Pictures (Images)

Speaker Notes:
My goal was to build a multi-modal AI system. Multi-modal means the AI doesn't just look at one thing. It acts like a real human detective by scanning three pillars: the behavioral statistics, the text in the bio, and the pixels in the profile picture.

Slide 4: The Tech Stack

Frontend (UI): HTML5, CSS3, JavaScript
Backend (API): Python, Flask
Machine Learning (AI): Scikit-Learn, Pandas, SciPy

Speaker Notes:

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To make this a real-world project, I didn't just train a model in a notebook. I built a Full-Stack application. I used HTML and CSS to clone an Instagram UI, Flask for the backend server, and Scikit-Learn to train the Machine Learning models.

Slide 5: Data & Feature Engineering

Feature Engineering: Extracting behaviors, not just data.
Numerical Features: Account Age, Follower/Following Ratio, Post Frequency.
Text Features: Bio keywords and sentence structure.
The Labels: Human (0) vs. Malicious Bot (1).

Speaker Notes:
A model is only as good as its data. I engineered specific features for the AI. For example, humans have a balanced follower-to-following ratio. Bots often follow 5,000 people but only have 10 followers. I extracted these behaviors so the AI could learn the patterns.

Slide 6: Algorithm 1 - Behavioral Analysis

Algorithm: Random Forest Classifier
Purpose: Scan numerical stats.
How it works: An ensemble of Decision Trees voting on anomalies.

Speaker Notes:
For the numerical data, I used a Random Forest Classifier. This algorithm works by creating a 'forest' of many Decision Trees. Each tree asks questions like 'Is the account age less than 5 days?' All the trees vote, and the majority wins. It is highly accurate for spotting unnatural behavior.

Slide 7: Algorithm 2 - Text Analysis (NLP)

Algorithm: Natural Language Processing (NLP)
Technique: TF-IDF (Term Frequency-Inverse Document Frequency)
Purpose: Convert text into mathematical vectors to find malicious language.

Speaker Notes:
Algorithms can't read English, so I used Natural Language Processing. Specifically, I used TF-IDF vectorization. This converts the words in a profile's bio into a mathematical matrix. It learns that words like 'free crypto' or 'click my link' have a high probability of being malicious.

Slide 8: Algorithm 3 - Image Analysis (Computer Vision)

Algorithm: Deep Learning / Computer Vision

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Purpose: Detect AI-generated 'Deepfake' profile pictures.
How it works: Analyzes unnatural pixel blending and asymmetrical lighting.

Speaker Notes:
Finally, to catch modern bots, I integrated Computer Vision. This scans the profile picture for Deepfakes. AI-generated faces from tools like Midjourney often have unnatural pixel smoothing on the skin or weird lighting. The vision algorithm catches what the human eye misses.

Slide 9: System Architecture

Step 1: User interacts with UI (Clicks Scan).
Step 2: JavaScript sends API request to Flask server.
Step 3: Flask runs data through ML models.
Step 4: JSON response updates UI (Red/Green Alert).

Speaker Notes:
Here is how the software pipeline works. When a user clicks 'Scan' on the website, it sends an API request to my Flask backend. Flask feeds the data into the Machine Learning models, calculates the threat level, and sends a JSON response back to update the screen in real-time.

Slide 10: Live Demonstration

Live Demo
(Switch screens to your VS Code and Web Browser here!)

Speaker Notes:
Theory is great, but let me show you how it actually works. I will now run the application...

Slide 11: Conclusion & Future Scope

Conclusion: Multi-modal AI is necessary for modern security.
Future Scope:
- Connect to the real Instagram API.
- Add a Chrome Extension version.
- Train on massive datasets like Kaggle's Twitter Bot dataset.

Speaker Notes:
In conclusion, catching bots requires looking at numbers, text, and images together. In the future, I would love to connect this directly to the Instagram API or turn it into a Google Chrome extension that protects users while they browse.

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Slide 12: Q&A

Thank You!
Questions?

Speaker Notes:
Thank you for your time and for reviewing my project. I am happy to answer any questions about the code or the Machine Learning theory!